

USER MANUAL · v0.6.1

BeatUp Engine

+ Silencer

Windows x64 · Chart Editor · Patcher · Themes · Silencer · BPM Workflow

GitHub	github.com/bishop
Website	web.beatupengine.workers.dev

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1. About the Project

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BeatUp Engine is a native Windows rhythm game focused on accurate timing, an authentic BeatUp / Audition-style keypad layout, chart editing and local patching. It is organised around two applications: **BeatUp Engine** itself and the separate **Silencer** utility for audio and timing preparation.

Component	Primary purpose
BeatUp Engine	Play songs, edit charts, review timing and import/export chart data.
Chart Editor	Create and adjust note charts on a beat-based timeline, review them with AutoPlay and split view.
Patcher	Game-facing mode with a song list, chances, Challenger Mode, results and rankings.
Theme Editor	Style the whole interface with six base colours, with a live preview.
Silencer	Inspect audio, refine BPM and offset, choose Start MAD I and verify timing visually.
BPM Detection Kit	Produce a starting BPM and offset estimate from song audio.

Core principle

BPM, Start MAD I and offset are timing data. Once they are stable, chart placement becomes reliable and long-range drift is much easier to diagnose.

2. Installation and File Structure

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The release is portable. Keep the executable and the **assets** folder together after extracting the archive.

1. Download the release archive from the BeatUp Engine page.
2. Extract it into a dedicated folder.
3. Keep the **assets** folder next to **BeatUpEngine.exe**.
4. Launch **BeatUpEngine.exe**. On first start, the onboarding appears (see chapter 3).

```
BeatUp Engine/
|-- BeatUpEngine.exe
|-- config.ini          (created on first start)
|-- theme.ini           (your saved themes)
|-- assets/
|   |-- arrows/ bars/ combo/ space/ sounds/ font/
|   |-- manual.pdf
|-- patcher/            (song list and patcher content)
```

File / folder	Content
config.ini	All settings: resolution, FPS, sounds, editor options, colours, creditor name.
theme.ini	Themes you saved in the Theme Editor.
assets/arrows	Arrow sprites, receptor outlines and the hit explosion.

File / folder	Content
assets/bars	Lane backgrounds and the chance icons.
assets/space	Space bar and Finish Move graphics.
assets/sounds	Gameplay sounds such as beat, miss and hover.
patcher/	Song list and song data used by the Patcher.

Updating

Replace the executable and keep your assets folder, **config.ini** and **theme.ini**. Your settings, themes and projects carry over. Themes saved with a version before v0.6 may need to be saved once more in the Theme Editor.

Requirements: Windows 10 / 11, 64-bit.

3. First Launch, Loading and Updates

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3.1 Onboarding

On the very first start, a short onboarding shows the most important hotkeys and asks for your **creditor name**. This name is written into every **.slk** you export. You can change it at any time in **Options** → **Editor**.

3.2 Loading screen

The loading screen works in a fixed order:

1. **Update check** — asks GitHub whether a newer version exists.
2. **Update notice** — if there is one, a single notice appears. **Open download page** opens the BeatUp Engine page, **Later** continues.
3. **File checks** — folders, core files and assets are verified, then the main menu opens.

The notice appears on every start while a newer version is available. Tick **Mute this message for 7 days** to hide it for a week. Without an internet connection the check simply finishes and the game starts normally.

Every loading screen, countdown and pause menu shows a random **tip**.

4. Main Menu

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The main menu greets you with **Welcome back, name** and offers four entries: **Editor**, **Patcher**, **Options** and **Exit**. Hovering an entry enlarges it slightly.

You can place your own background image as **assets/menu_background.png**. Without one, a colour gradient is shown.

Key	Function
Mouse	Select an entry.
Esc	Close Options, or ask whether to quit.
F11	Toggle borderless fullscreen.

5. Options

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Options are grouped into four tabs. All changes are saved when you press **Close**.

Tab	Setting	Description
Graphics	Frame Rate Limit	200, 500 or 1000 FPS.
	Logical Resolution	1024×768 up to 3840×2160. Resolutions larger than your monitor are disabled.
	Shadows	Darkens the whole window, from 0 to 85 %.
Game	Enter Lock	On by default. Keeps Enter-Chance active in Editor and Patcher; Numpad Enter no longer switches it off.
	Instant Play	The Patcher skips ahead to one MAD1 before the countdown.
	Startup Mode	<i>Patcher + Editor</i> , or <i>Editor Only</i> to start directly in the editor.
Editor	Auto Enter-Chance	Enter-Chance is switched on automatically when a song starts, but stays switchable.
	Creditor	The name written into every exported .slk.
	Theme	Quick switch between the built-in themes.
Patcher	Challenger Mode	Enables the Challenger rules (chapter 9).
	Default Chance	The chance selected when a Patcher song starts.

5.1 Resolution and fullscreen

The resolution decides the window mode: a resolution smaller than your monitor opens a movable window; a resolution equal to your monitor becomes borderless fullscreen. **F11** switches between your window size and fullscreen at any time.

6. Theme Editor

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Open it via **Options** → **Theme Editor...** in the editor menu. Instead of dozens of individual interface colours, you set **six base colours**. Everything else — hover states, sliders, checkboxes, separators — is derived automatically. Changes apply **live** to the whole interface, and a preview card shows buttons, a slider, a checkbox and text in the current style.

Colour	Used for
Background	Window background.
Surface	Cards, input fields and buttons at rest.
Accent	Active buttons, sliders, selection.
Text	Main text.
Muted	Hints and secondary text.
Border	Outlines and dividers.

6.1 Theme library

The left side lists the built-in themes (Mocha, Macchiato, Frappé, Latte) and every theme you saved in **theme.ini**. Click one to load it. Enter a name at the bottom and press **Save to theme.ini** to store your own theme. **Revert** returns to the state when the window was opened. Built-in themes cannot be overwritten.

6.2 Chart Editor colours

Setting	Description
Minimap notes / marker	Colour of the note bars and the position line in the minimap.
Waveform / core	Outer peak envelope and the inner line of the waveform.
Waveform opacity	Transparency of the waveform, 0 to 100 %.
Inner line	Shows a subtle core inside the waveform, similar to Arrow Vortex.
Position indicator	<i>Line</i> or <i>Receptor bar</i> (see 7.4).
Band / Icons / Edges	Colours of the receptor bar. <i>Icons</i> tints your asset graphics; white shows them unchanged.

The score bar and the result card also follow the current theme.

7. Chart Editor

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7.1 Views

Key	View
F1	Game view: the gameplay screen with the compact player in the top right.
F2	Chart view: the timeline only. The playback controls move into the menu bar so nothing covers the chart.
F3	Split view: chart on the left, game on the right.

In split view, drag the **divider** in the middle to give the chart or the game more room. The game view scales as a whole, so nothing is cut off. Double-click the divider to reset to 50/50.

7.2 Importing audio and projects

1. Import the song audio with **Ctrl + Alt + A** and enter BPM and offset.
2. Confirm that the audio start and the grid are aligned before placing many notes.
3. Save your work as a **.bu** project with **Ctrl + Shift + S**.

A **.bu** project stores the song, notes and your position. Double-clicking a **.bu** file opens it directly in the editor, and playback continues from the saved position. The editor always starts empty; music previewed in the Patcher is not carried over.

7.3 Columns and note placement

The chart columns are ordered **7 4 1 9 6 3 SP F**: the six arrows as in the game, then Space and Finish. The number keys match the columns:

Key	Column
1 / 2 / 3	Arrow 7 / 4 / 1 (left side)
4 / 5 / 6	Arrow 9 / 6 / 3 (right side)
7 or S	Space
8 or F	Finish Move

Each key toggles a note at the current beat: pressing it again removes the note. The grid supports 1/4, 1/8 and 1/16 snapping. Grid lines are colour-coded: white = MADI, red = 1/4, blue = 1/8, yellow = 1/16.

7.4 Position indicator

By default, the current position is shown as a **receptor bar** that mirrors the in-game receptors: the six arrow receptors, Space and Finish Move, each exactly in its column and at the same size as the notes passing over them. You can switch back to a plain line via **Options** → **Receptor Bar** or in the Theme Editor.

7.5 Player

The player shows play/pause, stop, the current time, a progress bar, the total time and the playback speed. Hovering the progress bar shows the time you would jump to; click or drag to seek. The speed is green above 100 %, neutral at 100 % and red below.

7.6 Info card

The info card in the top left shows BPM, beat, MADI, **Total Combo** and **Max. Score** — the score reached when every note is hit Perfect.

7.7 Editor controls

Key	Function
F5 / Space	Play or pause.
F6	Toggle AutoPlay (on by default in the editor).
F4	Toggle the metronome.
Arrow Up / Down	Jump one beat back / forward. With Shift: one MADL.
Mouse wheel	Scroll through the chart.
Ctrl + Left / Right	Change playback speed. Ctrl + 0 resets to 100 %.
Ctrl + drag	Select a range of notes, across both sides.
Ctrl + Z / Y	Undo / redo.
Ctrl + C / V / A	Copy, paste, select all.
Ctrl + W / E	Mirror / vertically mirror the selection.
Ctrl + B	Add a bookmark.
Delete	Delete the selection.
Tab	Toggle the tab selection.
Ctrl + S	Sound settings.
Ctrl + Shift + S	Save project (.bu).
Ctrl + Shift + E	Export .slk.
Ctrl + Shift + V	Export .ssc.
Esc	Clear the selection, then ask whether to save before leaving.

8. Gameplay and Controls

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BeatUp Engine uses the Audition / BeatUp keypad layout.

Key	Meaning
Num 7 / 4 / 1	Left side: top / middle / bottom.
Num 9 / 6 / 3	Right side: top / middle / bottom.
Space or Ctrl	Space input.
Num Enter	Toggle Enter-Chance (disabled while Enter Lock is on).
Esc	Open the pause menu.

8.1 Pause menu

Resume continues after a short “Get ready” countdown, **Restart Song** starts the song again and **Back to Song Selection** returns to the Patcher.

8.2 Judgment

Each key press is judged against the note it belongs to. The judgment windows, measured in beats from the exact hit point, are:

Judgment	Window
Perfect	within ± 0.18 beat
Great	up to ± 0.42 beat
Cool	up to ± 0.60 beat
Bad	beyond that, until the note is missed
Miss	the note passed without being hit

Timing diagnosis

A chart that is correct at the beginning but slowly drifts later in the song is usually a BPM problem, not an offset problem. Offset sets the starting position; BPM controls how the grid progresses over time.

9. Patcher

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The Patcher is the game-facing part: a song list, gameplay chances, results and rankings. It is separated from the editor so gameplay can be tested without the full authoring interface.

9.1 Chances

Open the gear menu to choose a chance for the session.

Chance	Behaviour
Default	Normal gameplay.
Rapid	Arrows travel at about 2× speed; the final quarter-beat returns to normal.
Vanishing	Arrows fade out smoothly as they approach the receptor.
Wave	A subtle sine-wave motion is applied to the arrows.
Inverted	Arrows are mirrored (1 <-> 9, 4 <-> 6, 7 <-> 3); input is inverted to match.
Speed Up	Starts at normal speed and increases towards 2×.
Halfway	Arrows appear from the midpoint of their path; timing stays normal.

During gameplay, **Ctrl + Alt + 1 to 7** switches the chance live in the order listed above.

9.2 Challenger Mode

With Challenger Mode enabled, the rules become active after **200 notes**. The run fails when the Perfect share drops below **80 %** or Cool + Bad + Miss together exceed **15 %**. A failed run shows **YOU FAILED** and always receives rank **F**, also in the leaderboard.

9.3 Song offset

Each song in the Patcher song list has its own offset. The standard value **0.7** plays with the engine's normal timing. A different value shifts the judgment slightly: a higher offset is judged later, a lower one earlier. The shift grows the slower a song is, as in the original game.

10. Results and Ranking

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10.1 Score bar

During play, a compact bar at the bottom shows score, combo and all five judgments with their share. A single thin bar underneath shows the distribution of your hits at a glance.

10.2 Result card

After the song, the result card shows the song, level and BPM, your **final score** and **max combo**, all judgments with their share and your **rank**. A line explains how the rank was calculated. Press **OK** or Enter to continue.

10.3 How the rank is calculated

1. **Base:** the share of Perfects, $\text{Perfect} \div (\text{Perfect} + \text{Cool} + \text{Bad} + \text{Miss})$. Great does not count — it neither lowers nor raises the result.
2. **Deductions:** 0.5 per Miss, 0.3 per Bad and 0.1 per Cool, in percentage points.
3. **Full Combo bonus:** +0.5 when there is no Miss and no Bad.
4. The result is rounded to one decimal and limited to 0–100 %.

Rank	Required
S+	98 % and above
S	95 % and above
A	90 % and above
B	85 % and above
C	80 % and above
D	75 % and above
E	70 % and above
F	below 70 %, or a failed Challenger run

Example

A Perfect share of 96.1 % with 4 Miss, 6 Bad and 23 Cool: $96.1 - 2.0 - 1.8 - 2.3 = 90.0 \%$ → rank A.

11. Chart Validation and Export

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11.1 Duplicate notes

The editor marks conflicting notes on the same beat in red. **S+F** and **N+F** are allowed combinations; other same-beat conflicts, including N+S, count as duplicates.

11.2 Export

Ctrl + Shift + E exports a **.slk**. Your creditor name from the settings is written into every export automatically. **Ctrl + Shift + V** exports an **.ssc**.

1. Resolve all red rows except the allowed combinations.
2. Review the chart with AutoPlay and in split view.
3. Export, then open the exported file again.
4. Check audio, timing, note order and credit.
5. Play the final chart in the Patcher.

12. Silencer

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Silencer is the companion utility for timing preparation. It combines waveform viewing, playback, MADi markers, Start MADi selection, offset control and BPM analysis. It supports WAV, MP3, OGG and FLAC.

Function	Purpose
Waveform view	Inspect the complete song on a vertical timeline and locate strong musical events.
MADi markers	4/4 measure references shown as M1, M2, M3 and so on.
Start MADi	Click a marker to choose the measure that defines the chart start.
Offset	Move the lead-in reference without changing the BPM. +1 / -1 Beat moves by exactly one beat.
Playback	Play, pause and stop; click the waveform to seek. The timeline follows playback.
Volume	Adjust the listening level without changing timing.

12.1 Recommended offset method

1. Estimate the BPM first.
2. Find the first strong musical beat.
3. Select a sensible Start MADi.
4. Apply the smallest offset needed to align the grid with the audio.
5. Check a later part of the song. If it drifts, reconsider the BPM instead of changing the offset again.

Example at 240 BPM

One beat is 0.250 seconds. An offset of -0.250 becomes 0.000 after +1 Beat, then 0.250 after another +1 Beat.

13. BPM Detection Kit

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The BPM Detection Kit gives a starting point for song timing. It evaluates onset and tempo candidates in an Arrow Vortex-style approach rather than guessing from the first peak.

Field	Meaning
BPM	Primary tempo estimate, shown to three decimals.
Offset	Estimated timing correction; negative values are valid.

Field	Meaning
Candidates	Alternative tempo candidates.
Format, sample rate, duration, bitrate	Basic audio file information.

Treat the result as a starting point. A wrong-but-plausible BPM can be more damaging than an obviously wrong one, because the whole chart drifts slowly. Always confirm the BPM over a long section in Silencer.

14. Recommended Chart Workflow

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Step	Action	Why it matters
1	Prepare the song	Keep the original audio as a backup.
2	Detect BPM	Get a BPM, an offset and alternative candidates.
3	Open in Silencer	Inspect duration and waveform before changing timing.
4	Set Start MADI and offset	Align the grid with the first strong beat.
5	Verify long-range timing	Drift later in the song usually means a wrong BPM.
6	Chart in the editor	Start coarse, then refine with 1/8 and 1/16.
7	Review	AutoPlay and split view reveal timing and pattern mistakes.
8	Resolve duplicates	Fix red rows except the allowed combinations.
9	Export and re-open	Verify the actual exported file.
10	Final test	Play the chart in the Patcher.

Creator mindset

Silencer solves timing. The editor solves chart construction. The Patcher solves final gameplay verification. Keeping them separate makes problems much easier to find.

15. Troubleshooting

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Problem	What to check
Chart looks shifted	Check Start MADl and offset in Silencer. Do not fix a wrong BPM with a large offset.
Timing drifts later	Re-check the BPM. Offset changes the start; BPM controls the slope.
Arrows appear as coloured bars in the editor	An arrow graphic is missing. Make sure the complete assets folder is next to the executable.
A judgment sound does not play	Check the file chosen in the sound settings (Ctrl + S) and that it exists in assets/sounds.
Custom asset not visible	Check folder and file name, then restart the application.
Saved theme does not appear	Themes saved before v0.6 may be damaged. Save them once more in the Theme Editor.
Update notice keeps appearing	Update to the latest version, or tick "Mute this message for 7 days".
Numpad Enter does nothing	Enter Lock is on. Turn it off in Options → Game.
Everything looks dark	Check Options → Graphics → Shadows.
Troubleshooting rule Change one thing at a time. When diagnosing timing, never change BPM, Start MADl and offset at the same time.	

16. Quick Reference

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Task	How
Toggle fullscreen	F11
Switch views	F1 game · F2 chart · F3 split
Place notes	1-6 arrows · 7 / S Space · 8 / F Finish
Navigate the chart	Arrow Up / Down (Shift = one MADI) · mouse wheel
Play / pause	F5 or Space
AutoPlay / metronome	F6 / F4
Playback speed	Ctrl + Left / Right · Ctrl + 0 resets
Save project	Ctrl + Shift + S
Export .slk / .ssc	Ctrl + Shift + E / Ctrl + Shift + V
Import audio	Ctrl + Alt + A
Sound settings	Ctrl + S
Change chance live	Ctrl + Alt + 1-7 (Patcher)
Style the interface	Options → Theme Editor
Refine timing	Silencer

Links: github.com/bishpop · web.beatupengine.workers.dev

Manual scope

This manual describes BeatUp Engine v0.6.1. Option names or file locations may change in future builds; the shipped interface is then authoritative.